

Use Case And Business Understanding

Business Intelligence Analysis of Public Transportation Delays

I. Industry :

The project relates to the public transportation sector. Public transport plays a key role in daily mobility, especially for students, workers, and low-income citizens. However, this sector faces several challenges such as service delays, overcrowding during peak hours, and unequal transport availability between regions.

Efficient management of public transport is essential to ensure smooth commuting, optimize resource allocation, and improve citizen satisfaction.

II. Organization Description :

This Business Intelligence solution is designed for a public transportation authority responsible for managing transport services such as buses and metro systems.

- The organization plans routes, manages vehicles and staff, and ensures that transport services are available to meet citizens' needs.
- The BI system will help transport managers and planners by providing analytical insights to improve service efficiency, monitor demand, and optimize route planning.

III. Business Problem :

Public transportation often faces challenges such as delays, overcrowding during peak hours, and uneven service coverage across different areas.

Decision-makers currently rely on basic reports and fragmented statistics, which makes it difficult to identify inefficiencies or plan improvements.

Without a centralized system, it is hard to monitor transport performance, compare routes and regions, detect peak demand periods, or ensure services meet citizens' needs.

The main problem is the lack of a unified Business Intelligence system that transforms transport related data into meaningful insights to support operational and strategic decisions.

The Problem statement :

The public transportation authority lacks real-time and historical visibility into public transport operations (delays, frequency, capacity, punctuality). This results in poor service

reliability, inefficient scheduling, overcrowding, and low customer satisfaction.

IV. Project Objective :

The objective of this project is to design a Business Intelligence dashboard that analyzes public transport activity and provides actionable insights to improve service quality.

The dashboard will help transport managers and planners to:

- Monitor transport demand trends over time
- Identify peak usage periods and overcrowded routes
- Compare service coverage and efficiency across different areas
- Support better planning and optimization of transport resources

By consolidating transport data, analyzing performance metrics, and presenting insights through interactive visualizations, the dashboard enables decision-makers to detect inefficiencies, optimize resource allocation, and improve overall transport performance.

V. Analytical questions :

1. What is the average delay per transport line?
2. Which stations experience the highest delay frequency?
3. How does delay vary by time of day (peak vs off-peak)?
4. What percentage of trips arrive on time?
5. Which days of the week have the worst service reliability?
6. Is there a relationship between passenger volume and delays?
7. Which routes suffer from overcrowding?
8. How do planned vs actual arrival times differ?
9. Which routes and time periods experience the highest delays?
10. Which lines require operational improvement first?

VI. Key Performance Indicators :

- **Average Delay (minutes)** – Mean delay time per line, route, or station.
- **On-Time Arrival Rate (%)** – Percentage of trips arriving on schedule.
- **Trips Delayed > 5 minutes (%)** – Percentage of trips delayed more than 5 minutes.
- **Delay Frequency per Station** – Number of trips delayed more than 5 minutes at each station.
- **Trips per Route** – Total number of trips conducted on each route over a specific period.
- **Routes with Highest Delay** – Routes with the highest average delays.
- **Cancelled Trips Count** – Total number of trips that were cancelled or not completed.

- **Peak Delay Frequency (%)** – Percentage of trips delayed more than 5 minutes during peak hours.

Wejdene Chouchene

Thana Errahmen Meassoudi

Ghofrane Ghouel

Youssef Harbi

BA/IT